

OSI Model and TCP/IP

Reference Models: Standardized frameworks for network communication.

OSI Model

7 Layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.

TCP/IP Model

4 Layers: Network Access, Internet, Transport, Application.

Encapsulation

Process of adding headers to data as it moves down the layers.

Physical Layer:

Deals with transmission of raw bits over a communication channel.

Transport Layer:

Process-to-process delivery, segmentation, reliable/unreliable delivery (TCP/UDP).

Data Link Layer:

Node-to-node delivery, framing, error control, flow control, MAC.

Application Layer:

Provides network services to user applications (HTTP, FTP, SMTP).

Network Layer:

Source-to-destination delivery of packets, routing, logical addressing (IP).